

Allatostatin-C-like Receptor Analysis

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A0A891XI80

First Domain

Sequence [52-342]: PTLLLITQVLCAAVCIVGLLGNSLVIYVVLRFQSKMQTVTNMYIVNLAVADECFLLIGVPP
LLATMNLRIWPFQSIMCKAYMTTTSINQFTSSIFLTIMSADRYIAVCHPISAPKVRTPLISKVVSASAWTASALLMIP
IFIYANIMTNENVVSCNLTWPESENVSSQGAFTLYTFILGFSIPLMLISVFYILVIRKLKTVGPKNKSKEKKKSHRKV
TKLVLTVITVYVLCWLPYWLTQMALIFTPPKQCQSRVSITVSLLAGCLSYSNSAMNPILYAFLSDNFKKSFLLKACT

Homologies

Psi-blast search against Swiss-Prot:

First 5 Full Homologues

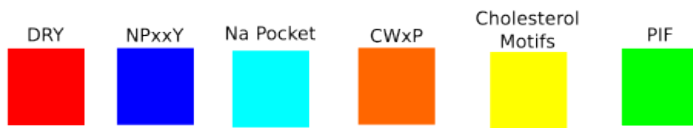
##	receptorNames	identities	positives	organisms
## 1	Somatostatin receptor type 2	47%	68%	Homo sapiens
## 2	Somatostatin receptor type 1	43%	68%	Rattus norvegicus
## 3	Somatostatin receptor type 1	43%	68%	Mus musculus
## 4	Somatostatin receptor type 1	43%	68%	Homo sapiens
## 5	Somatostatin receptor type 1	43%	68%	Canis lupus familiaris

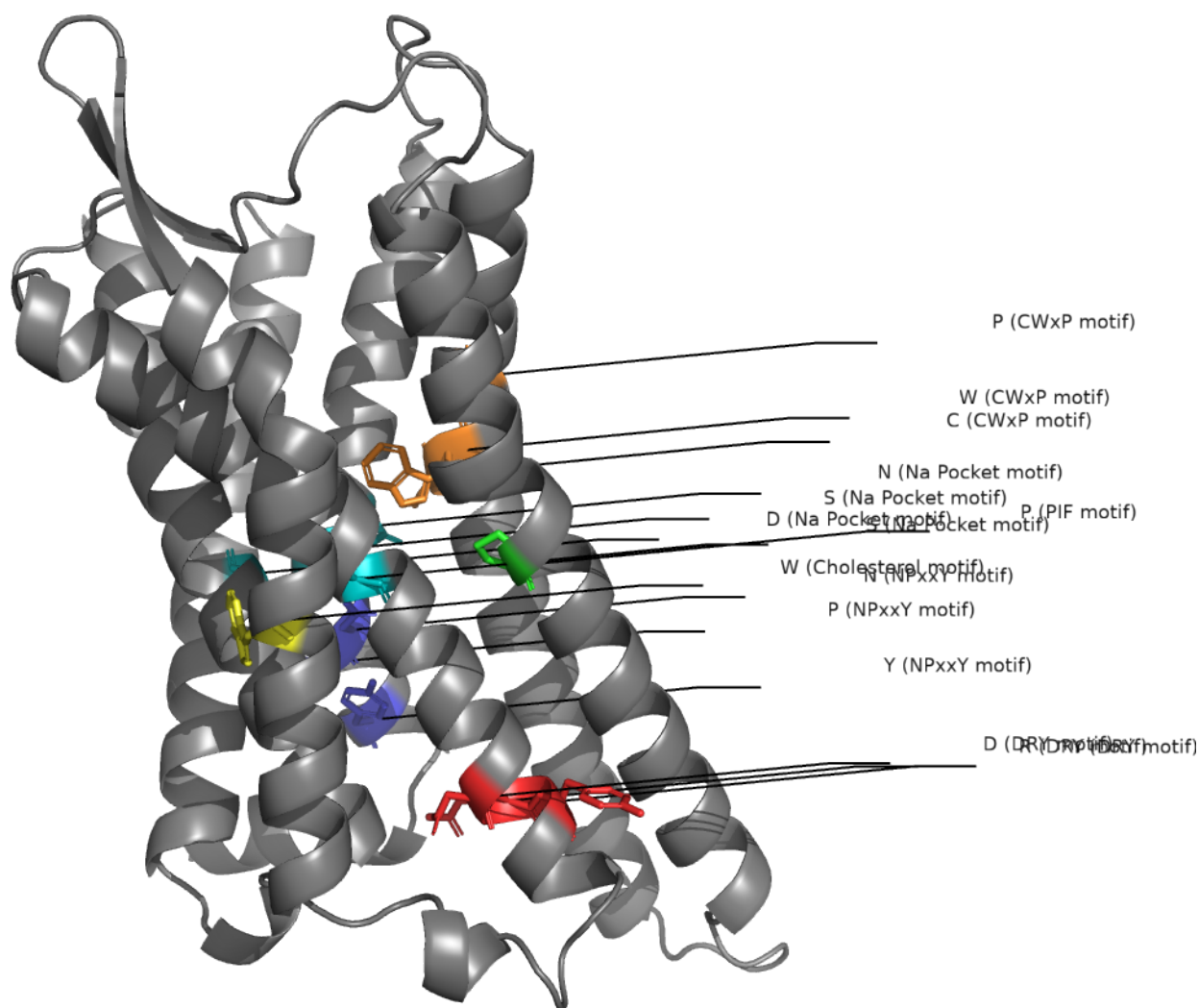
First 50 Unique Homologues

## [1]	"Somatostatin receptor type 2"	"Somatostatin receptor type 1"
## [3]	"Somatostatin receptor type 4"	"Somatostatin receptor type 5"
## [5]	"Somatostatin receptor type 3"	"Nociceptin receptor"
## [7]	"Kappa-type opioid receptor"	"Neuropeptides B/W receptor type 2"
## [9]	"Delta-type opioid receptor"	"Mu-type opioid receptor"
## [11]	"Neuropeptides B/W receptor type 1"	"Galanin receptor"

The similarity of the 50th match: 33% identity, 57% positives

Structure Motifs





Conservation

Conservation of the protein from SwissProt with highest homology to the first domain: Somatostatin receptor type 2, UniProt ID: P30874

Second domain

Sequence [448-700]: TNLLIINLALADLLFIVFCVPFTATDYMLPYWPFGDVWCKIVQYLIVVTAYASVYTLV
LMSLDRFLAVVHPIASISVRTERNAVTAIGVTWVILLSSVPVYL SHGVTNYIYLSSEHTACVFLDDYNKPVFQILFF
ATSYVIPLALICGLYLCMLMRLWRGVAPGGHVSAESRRGRKRVTMRMVVVVVAIFAVCWFPQMILVLKSVDREYITNT
SVMVQIVSHVLAYMNSCVNPILYAFLSENFRKA FRKVI

Homologies

Psi-blast search against Swiss-Prot:

First 5 Full Homologues

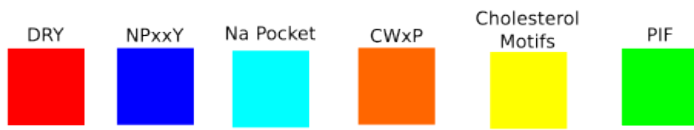
##	receptorNames	identities	positives	organisms
## 1	Allatostatin-A receptor	70%	85%	Bombyx mori
## 2	Galanin receptor 2a	40%	63%	Danio rerio
## 3	Galanin receptor type 2	40%	61%	Mus musculus
## 4	Galanin receptor 2b	39%	62%	Danio rerio
## 5	Somatostatin receptor type 1	40%	61%	Rattus norvegicus

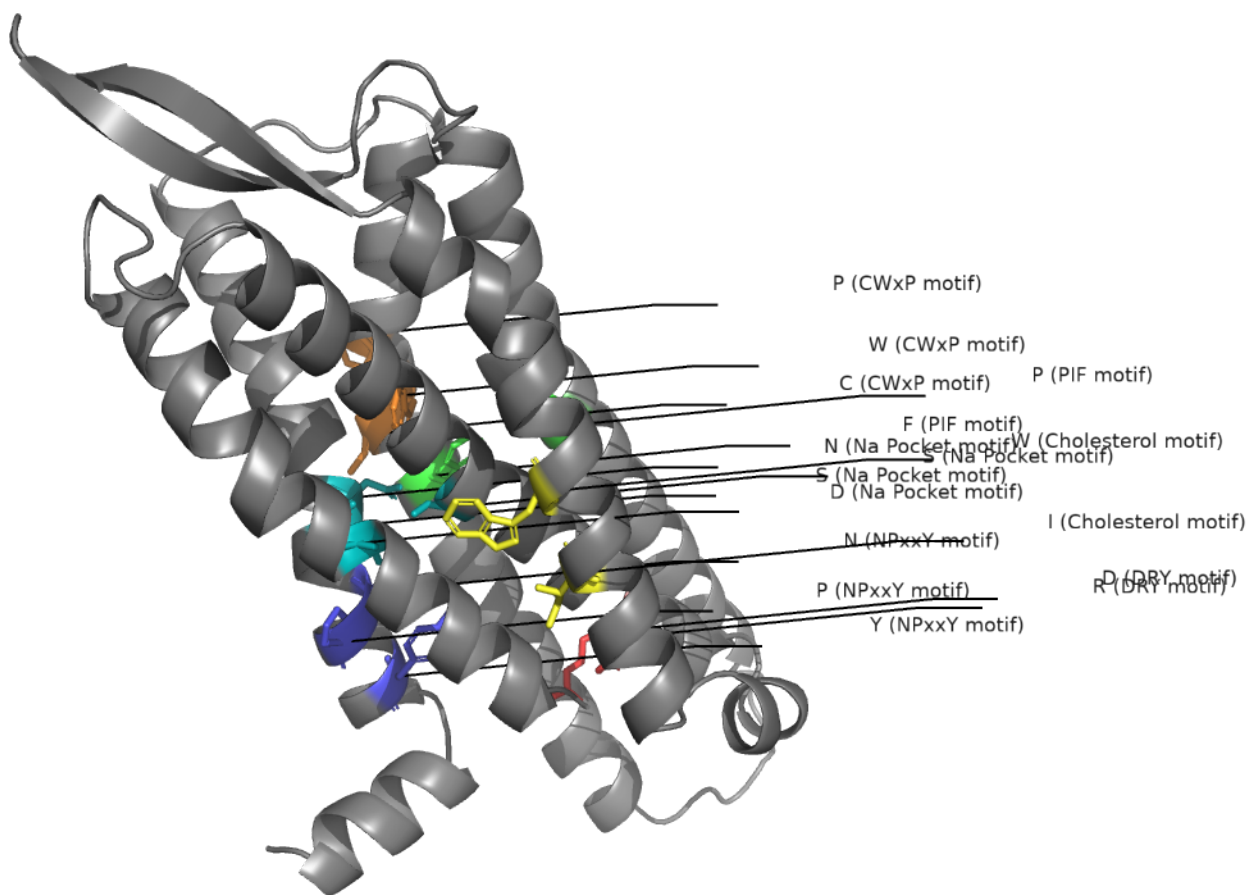
First 50 Unique Homologues

```
## [1] "Allatostatin-A receptor"
## [2] "Galanin receptor 2a"
## [3] "Galanin receptor type 2"
## [4] "Galanin receptor 2b"
## [5] "Somatostatin receptor type 1"
## [6] "Galanin receptor type 3"
## [7] "Galanin receptor type 1"
## [8] "Nociceptin receptor"
## [9] "Somatostatin receptor type 5"
## [10] "Mu-type opioid receptor"
## [11] "KiSS-1 receptor"
## [12] "Kappa-type opioid receptor"
## [13] "Somatostatin receptor type 2"
## [14] "G-protein coupled receptor 54 (orphan)"
## [15] "Somatostatin receptor type 3"
## [16] "Somatostatin receptor type 4"
## [17] "Delta-type opioid receptor"
```

The similarity of the 50th match: 35% identity, 53% positives

Structure Motifs





Conservation

Conservation of the protein from SwissProt with highest homology to the second domain: Allatostatin-A receptor, UniProt ID: Q8WPA2

Relative domains

Pairwise alignment

Genomic sequence